

David Adrian

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Education

University of Michigan, Ann Arbor, MI
PhD, Computer Science 2019

- ADVISOR: J. Alex Halderman
- RESEARCH FOCUS: Computer Security
- DISSERTATION: Using Large-Scale Empirical Methods to Understand Fragile Cryptographic Ecosystems

Masters of Science in Engineering, Computer Science Class of 2016

Bachelors of Science in Engineering, Computer Science Class of 2014

- MINOR: Mathematics

Work Experience

Senior Product Manager, Google 2022 - Present

- Chrome Security team, lead for secure connections, memory safety, and web platform security.
- Product owner for HTTPS certificates in Chrome, including the removal of the lock icon, deprecating plaintext HTTP, enterprise certificate management, and migrating to post-quantum cryptography.
- Product owner for all memory safety initiatives from the Chrome Security team, including expanding the use of Rust.
- Product owner for all cryptographic compliance needs in Chrome, including CNSA2.0.
- Product owner for web platform security, including rolling out the Local Network Access (LNA) permission

Principal Engineer, Nametag 2020 - 2022

- Nametag mediates PII between users and companies via a push notification approval flow (Stripe crossed with Duo for PII).
- Second employee, first non-executive hire.
- Backend service engineering and design, primarily in Go, with a focus and security, privacy, and anti-abuse.
- Led all cryptography engineering, including authorization token design and deployment, and mobile TPM key architectures.
- Built and defined core product features, including the user flows for multi-device support and automatic ID-based account recovery.

Cofounder, Censys 2017 - 2020

- Founded Censys as a research project to expose Internet-wide measurement data during my PhD at the University of Michigan.
- Cofounded a startup to commercialize the Censys research project into an enterprise data offering and SaaS security product.
- Censys collected scan data on over 4B hosts, processed over 2B X.509 certificates, and provided terabytes of data to enterprise customers daily.
- Set company-wide engineering direction and vision as Censys grew to over 50 employees and over \$3MM ARR, and a \$15M Series A.
- Launched the initial Censys attack-surface management SaaS application.
- Defined technical product direction, including how our SaaS application and data offering interact.
- Designed and implemented service architecture supporting multiple data-driven products, data warehousing, and ETLs.
- Migrated data infrastructure from on-campus (University of Michigan datacenter) to Google Cloud and Kubernetes.

	Software Engineering Intern, <i>Google Chrome Security</i> Summer 2016 • Worked on the Chrome TLS and X.509 stack • Implemented OCSP Expect-Staple in order to measure the feasibility of OCSP Must-Staple.
	Software Engineering Intern, <i>Duo Security</i> Summer 2013 • Two-factor authentication-as-a-service, using mobile phones as the second authentication factor and push notifications for login approval. • Grew various Python backend subsystems to support new features; handled all bugs, feature requests, and security considerations for the Duo Unix client. • Wrote a patch for OpenSSH to securely allow third-party authentication plugins.
Research Experience	Research Scientist and Collaborator, <i>Stanford Empirical Security Research Group</i> 2020 - Present • Work with Professor Zakir Durumeric at Stanford Computer Science. • Focus on secure transports and secure protocol design and development. • Part-time research scientist 2020-2022 (paid) PhD, <i>with Professor J. Alex Halderman</i> 2013 - 2019 • Computer security and Internet measurement research, concentrating on using global perspectives to gain insight into networks and cryptography. • Created and maintain ZGrab, ZCrypto, and ZMap, open-source tools for high-speed Internet-wide measurement (available on GitHub). • Maintain TLS and X.509 implementations, including certificate chaining, designed for measurement of the HTTPS ecosystem. • Released Censys, a search engine for Internet-wide measurement data. • Codiscovered the Logjam and DROWN attacks on TLS. Whisper Project, <i>with Professor Robert P. Dick</i> 2011 - 2013 • Released MANES, an Android prototyping framework for mobile ad-hoc networks • Released Shout, a user-facing censorship-resistant communication application built using MANES
Teaching Experience	Lecturer, <i>EECS 388: Intro to Computer Security</i> Fall 2016 Graduate Student Instructor, <i>EECS 388: Intro to Computer Security</i> Fall 2015 Graduate Student Instructor, <i>EECS 588: Computer Security</i> Winter 2015 Instructional Aide, <i>EECS 281: Data Structures and Algorithms</i> Winter 2014 Instructional Aide, <i>EECS 280: Programming and Data Structures</i> Fall 2013
Other Experience	Network Technician, <i>CAEN</i> 2011 - 2012 • Maintained over 700 wireless access points for the University of Michigan College of Engineering campus. • Transitioned printing system from CAEN control to campus-wide IT department. • Diagnosed and performed network troubleshooting as needed. • Configured L2 access on managed switches from Cisco, Juniper, and Force10.
Computer Skills	Languages: C, C++, Go, Java, Javascript, Python, Rust, Swift, Typescript Platforms/Frameworks: Android, Celery, iOS, Pylons, Node.js, POSIX, Pyramid, React (Native) Datastores: Bigtable, Elasticsearch, Kafka, MySQL, MongoDB, Postgres, Redis, RocksDB Data Processing: Apache Airflow, Apache Beam, Bigquery, Google Cloud Dataflow, Jupyter Infrastructure: Amazon (AWS), Bazel, Cloudformation, Docker, Google Cloud (GCP), Grafana, Kubernetes, Prometheus, Salt

Project Management: Agile, Asana, Confluence, Github, Gitlab, JIRA, Kanban, Notion
AI: Cursor, OpenAI Codex

**Conference
Publications**

Ten Years of ZMap

Zakir Durumeric, David Adrian, Phillip Stephens, Eric Wustrow, and J. Alex Halderman

ACM Internet Measurement Conference (IMC), 2024.

ZDNS: A Fast DNS Toolkit for Internet Measurement

Liz Izhikevich, Gautam Akiwate, Briana Berger, Spencer Drakontaidis, Anna Ascherman, Paul Pearce, David Adrian, and Zakir Durumeric

ACM Internet Measurement Conference (IMC), 2022.

On the Origin of Scanning: The Impact of Location on Internet-Wide Scans

Gerry Wan, Liz Izhikevich, David Adrian, Katsunari Yoshioka, Ralph Holz, Christian Rossow, and Zakir Durumeric

ACM Internet Measurement Conference (IMC), 2020.

Tracking Certificate Misissuance in the Wild

Deepak Kumar, Zhengping Wang, Matthew Hyder, Joseph Dickinson, Gabrielle Beck, David Adrian, Zakir Durumeric, J. Alex Halderman, and Michael Bailey

IEEE Symposium on Security and Privacy (Oakland), 2018.

Measuring small subgroup attacks against Diffie-Hellman

Luke Valenta, David Adrian, Antonio Sanso, Shaanan Cohney, Joshua Fried, Marcella Hastings, J. Alex Halderman, and Nadia Heninger

Network and Distributed System Security Symposium (NDSS), 2017.

An Internet-Wide View of ICS Devices

Ariana Mirian, Zane Ma, David Adrian, Matthew Tischer, Thasphon Chuenchujit, Tim Yardley, Robin Bertheier, Josh Mason, Zakir Durumeric, J. Alex Halderman, and Michael Bailey

IEEE Conference on Privacy, Security, and Trust (PST), 2016.

DROWN: Breaking TLS Using SSLv2

Nimrod Aviran, Sebastian Schinzel, Juraj Somorovsky, Nadia Heninger, Maik Dankel, Jens Steube, Luke Valenta, David Adrian, J. Alex Halderman, Viktor Dukhovni, Emilia Käspér, Shaanan Cohney, Susanne Engels, Christof Paar, and Yuval Shavitt

USENIX Security Symposium, 2016.

Neither Snow Nor Rain Nor MITM... An Empirical Analysis of Email Delivery Security

Zakir Durumeric, David Adrian, Ariana Mirian, James Kasten, Kurt Thomas, Vijay Eranti, Nicholas Lidzborski, Elie Bursztein, Michael Bailey, and J. Alex Halderman

ACM Internet Measurement Conference (IMC), 2015.

Imperfect Forward Secrecy: How Diffie-Hellman Fails in Practice

David Adrian, Karthikeyan Bhargavan, Zakir Durumeric, Pierrick Gaudry, Matthew Green, J. Alex Halderman, Nadia Heninger, Drew Springall, Emmanuel Thomé, Luke Valenta, Benjamin VanderSloot, Eric Wustrow, Santiago Zanella-Béguelin, and Paul Zimmermann

ACM Conference on Computer and Communications Security (CCS), 2015.

A Search-Engine Backed by Internet-Wide Scanning

Zakir Durumeric, David Adrian, Ariana Mirian, Michael Bailey, and J. Alex Halderman

ACM Conference on Computer and Communications Security (CCS), 2015.

Performance and Energy Consumption Analysis of a Delay-Tolerant Network for Censorship-Resistant Communication

Yue Liu, David R. Bild, David Adrian, Gulshan Singh, Robert P. Dick, Dan S. Wallach, and Z. Morley Mao

ACM Symposium on Mobile Ad Hoc Networking and Computing (MobiHoc), 2015.

The Matter of Heartbleed

Zakir Durumeric, Frank Li, James Kasten, Johanna Amann, Jethro Beekman, Mathias Payer, Nicolas Weaver, David Adrian, Vern Paxson, and Michael Bailey

ACM Internet Measurement Conference (IMC), 2014.

**Workshop
Publications**

Zipper ZMap: Internet-Wide Scanning at 10 Gbps

David Adrian, Zakir Durumeric, Gulshan Singh and J. Alex Halderman

USENIX Workshop on Offensive Technologies (WOOT), 2014.